Machine Learning

Getting your Kaggle data — Lecture 23 (Chapter 9, Part 4)

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Part 4 — Getting your Kaggle data

Today: your datasets

- Everyone has picked a competition. We'll open some of them together and read the pages
- Then: getting the data onto your computer, by hand and by command

Reading a competition page

- **Overview:** what is one example? What is predicted? Which **metric** scores you (accuracy, AUC, log-loss)?
- **Data:** which files, how big, what format? Which kind: one simple table, **several tables** linked by IDs, or per-moment data (time series, game-play capture) that must be summarized per example?
- **Rules:** you must **join** (accept the rules) before you can download. Does it still take **late submissions**?
- **Leaderboard:** what do the best get? That's your reference, not your target
- **Code:** other people's public notebooks. Read them to learn; don't copy them

Example competitions: good first projects (1/2)

#	Competition	Predict	Scored by	MB	Rows × features	Positive	Kind
1	Reservation Cancellation	hotel booking cancellations	AUC	4	—	—	simple
2	Kidney Stone Prediction	kidney stones from urine tests (small)	AUC	<1	—	—	simple
3	Software Defects	defective code modules	AUC	19	101,763 × 21	23%	simple
4	Smoker Status from Bio-Signals	smoking status	AUC	23	—	—	simple
5	Bank Churn	which customers leave the bank	AUC	22	165,034 × 12	21%	simple
6	Poisonous Mushrooms	edible or poisonous	MCC	298	—	—	simple
7	Loan Approval	loan approval	AUC	6	58,645 × 11	14%	simple

MB: all the competition's files. Rows × features and the share of the rarer class: from the training file, where already checked; — means join the competition and look. Click a name to open its Kaggle page. Checked with Kaggle's API, 4 November 2026.

Example competitions: good first projects (2/2)

#	Competition	Predict	Scored by	MB	Rows × features	Positive	Kind
8	Mental Health	depression from survey answers	accuracy	28	—	—	simple
9	Rainfall	rain tomorrow, from weather measurements	AUC	<1	—	—	simple
10	Introverts and Extroverts	personality type	accuracy	1	—	—	simple
11	Bank Marketing	who subscribes to a term deposit	AUC	90	—	—	simple
12	Insurance Cross Selling	interest in car insurance (a very large file: sample it)	AUC	1,196	—	—	simple

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Example competitions: more than two classes

#	Competition	Predict	Scored by	MB	Rows × features	Positive	Kind
13	Cirrhosis Outcomes	patient outcome, 3 classes	log-loss	1	7,905 × 18		simple
14	Obesity Risk	obesity level, 7 classes	accuracy	5	—	—	simple
15	Academic Success	graduate, enrolled or dropout	accuracy	16	—	—	simple
16	Steel Plate Defects	7 defect types (several can be true at once)	mean AUC per class	6	—	—	simple
17	Optimal Fertilizers	fertilizer name; scored on the top-3 guesses	MAP@3	50	—	—	simple
18	Forest Cover Type	forest type from map variables, 7 classes	accuracy	97	—	—	simple
19	Otto Group Products	product category, 9 classes, anonymous features	log-loss	44	—	—	simple

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Example competitions: more challenging (1/2)

#	Competition	Predict	Scored by	MB	Rows × features	Positive	Kind
20	Give Me Some Credit	serious financial distress in the next two years	AUC	14	—	—	simple (+ data dictionary)
21	Santander Customer Satisfaction	unhappy customers (imbalanced)	AUC	119	76,020 × 369	4%	simple
22	Santander Customer Transaction	a future transaction, from 200 anonymous features	AUC	606	—	—	simple
23	Porto Seguro Safe Driver	an insurance claim next year (imbalanced)	Gini = 2 AUC − 1	301	—	—	simple
24	BNP Paribas Cardif Claims	claims that can be fast-tracked	log-loss	104	—	—	simple
25	ICR: Age-Related Conditions	medical conditions from anonymous measurements (small, imbalanced)	balanced log-loss	<1	—	—	simple + metadata table
26	Tabular Playground May 2022	a synthetic manufacturing-control state	AUC	574	—	—	simple

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Example competitions: more challenging (2/2)

#	Competition	Predict	Scored by	MB	Rows × features	Positive	Kind
27	Tabular Playground Aug 2022	product failures	AUC	7	—	—	simple
28	Home Credit Default Risk	loan repayment; several tables to join (advanced)	AUC	2,684	—	—	8 tables to join
29	IEEE-CIS Fraud Detection	fraudulent transactions; large, two tables (advanced)	AUC	1,355	—	—	2 tables to join

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Three ways to get the data

- **Browser:** the Download button on the Data tab. Fine for this project
- **A command in your notebook:** reproducible, so you, or anyone, can get the data again
- **The Kaggle API:** a **REST API** (requests are URLs, replies are data) with a **command-line tool**, `kaggle`, and a Python library
- On **WSL** your browser saves to Windows. From Linux the Windows disk is under `/mnt/c/`:

```
cp /mnt/c/Users/<you>/Downloads/<competition>.zip ~/<your-project>/
```

The Kaggle command-line tool

```
# kaggle 2.2 also needs six on Python 3.13
pip install kaggle six
kaggle auth login --no-launch-browser
```

- `login` prints a link: open it, sign in to Kaggle, paste back the code it shows
- Or make an **API token** at kaggle.com/settings/api, and `export KAGGLE_API_TOKEN=...` or save it in `~/.kaggle/access_token`
- A token is a **password**: never put it in a notebook, a commit or a screenshot

Commands you'll use

```
kaggle competitions list -s mushroom      # search
C=playground-series-s4e8                  # the competition's name
kaggle competitions files $C              # what's there
kaggle competitions download $C -p data --unzip
kaggle competitions submit $C -f submission.csv -m "first try"
kaggle competitions submissions $C        # your scores
```

- The name is the end of the competition's URL: kaggle.com/competitions/**playground-series-s4e8**
- **403 Forbidden** means you haven't accepted the competition's rules: join it on the website first
- A dataset rather than a competition: `kaggle datasets download <owner>/<name> -p data --unzip`

In your notebook

```
!kaggle competitions download playground-series-s4e8 -p data --unzip
!ls -lh data
!head -3 data/train.csv
import pandas as pd
df = pd.read_csv("data/train.csv")
```

- Look before you load: the header row, the separator, the number of rows (`wc -l`), the size on disk
- **Keep the data out of Git**: add `data/` to `.gitignore`. Your README says where the data comes from, and the command gets it

The submission

- `test.csv` has the inputs and an **ID** column, but no answers. Keep the IDs: the submission needs them
- Predict every row of `test.csv`, write it in `sample_submission.csv`'s format, and submit
- Kaggle scores it against answers only it has. Late submissions get a score but no rank
- Before predicting, retrain your chosen model on all of `train.csv`

Vocabulary

- **competition, leaderboard, metric, rules, late submission, public notebook**
- **REST API, command-line interface (CLI), API token, OAuth login**
- **WSL, `/mnt/c/`, `.gitignore`, reproducible download**
- **`train.csv`, `test.csv`, `sample_submission.csv`, ID column, submission**